Introduction to Urban Data Science

CRP 4680/5680 Spring 2025

Week 12 Machine Learning (I)

Wenzheng Li

Announcement

- Assignment 4 is due April 20th.
- My Office hours next week: Monday (April 14) from 2:30pm – 6:30pm
- Guest speaker on Thursday (April 17):
  - Waishan Qiu from the University of Hongkong;
  - starting at 9am via zoom.

OUTLINE

- Introduction to Machine Learning
  - What is machine learning?
  - Machine learning types
- Supervised Learning
  - Classification and Regression
  - Understand machine learning
  - Machine learning vs. statistics

1 What is Machine Learning

What is Machine Learning?

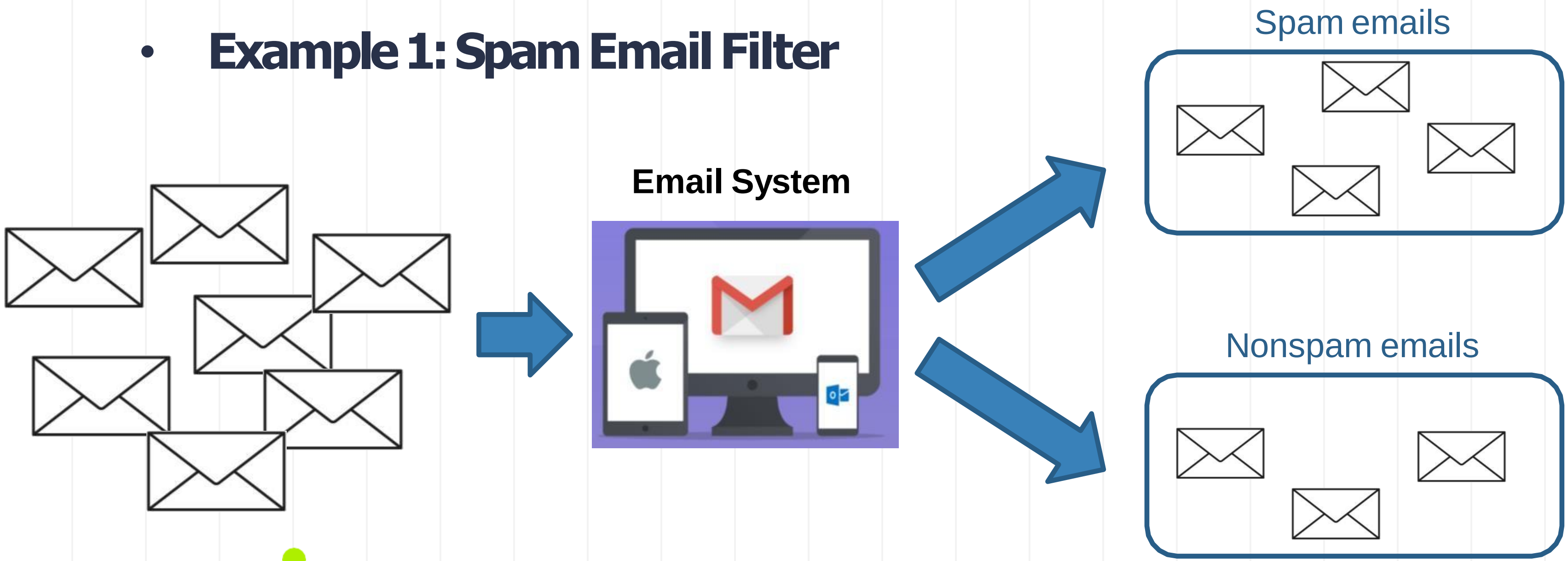
[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

—Arthur Samuel, 1959

Samuel, Arthur (1959). "Some Studies in Machine Learning Using the Game of Checkers". IBM Journal of Research and Development. ***3*** *(3): 210–229.*

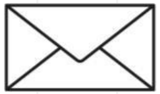
What is Machine Learning?

- **Example 1: Spam Email Filter**



What would you do to create such a system (machine)?

- **Task**: filter spam emails



```
if conditions:
    it is a spam
else:
    it is not a spam
```

Spam emails

Nonspam emails

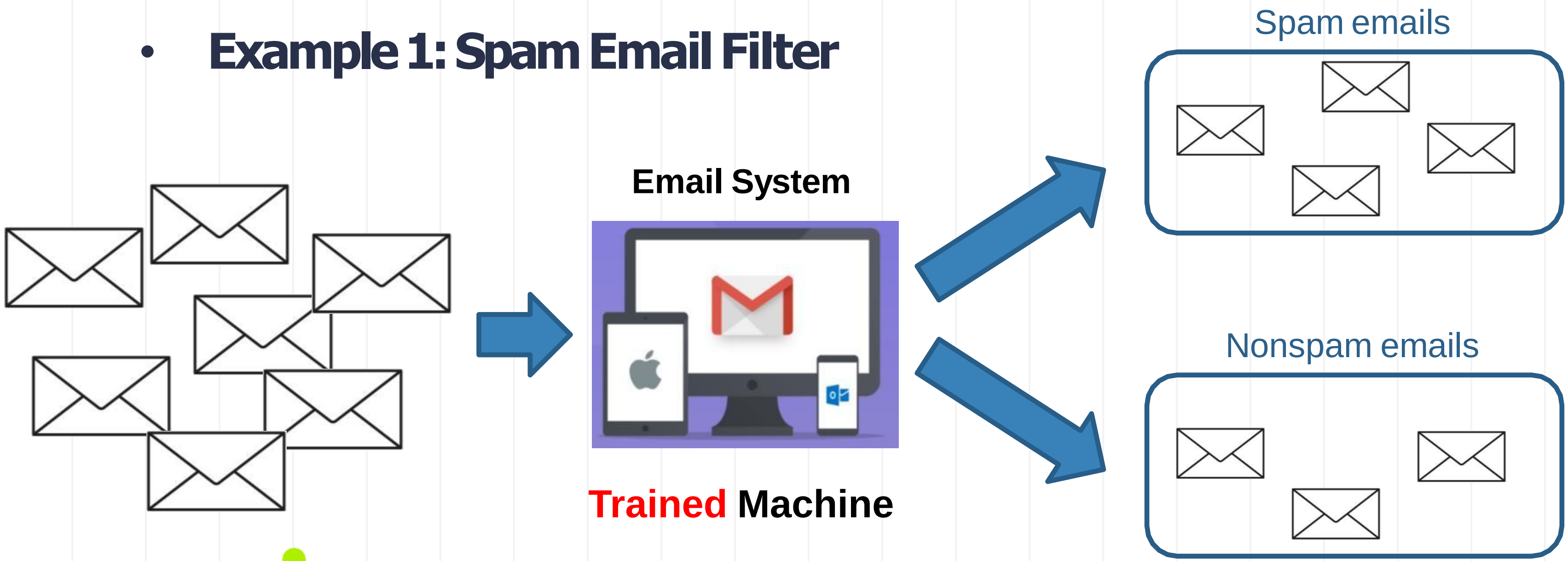
What are the ***conditions***?

- bad words
- ads (what are the ads)?
- suspicious senders?
- in title? In email body?

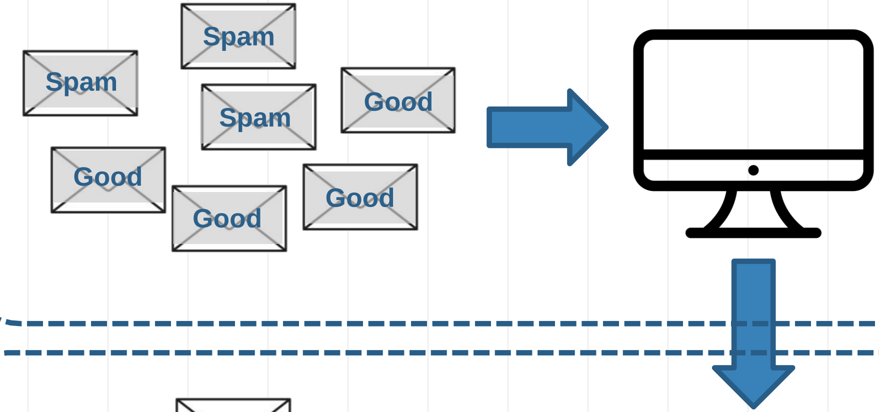
Not possible to write a rule to exhaust all cases!

What is Machine Learning?

- **Example 1: Spam Email Filter**



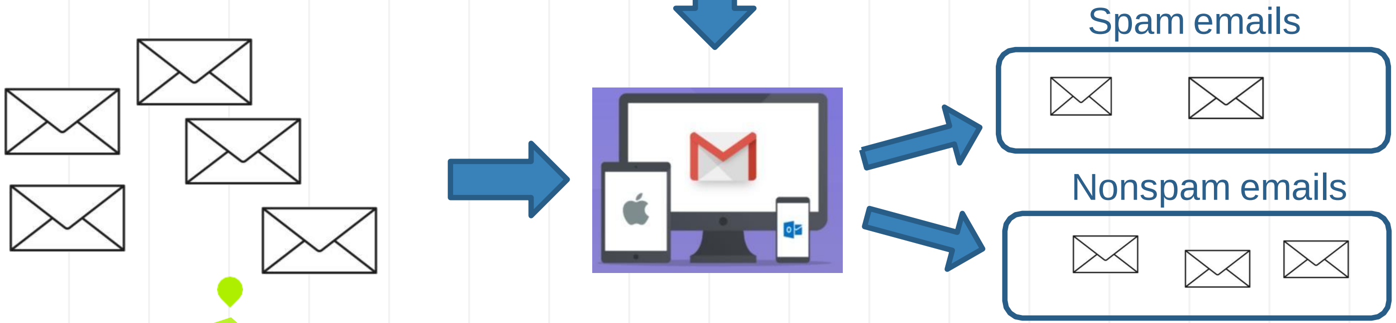
A span filter based on machine learning



conditions:

- bad words
- ads (what are the ads)?
- suspicious senders?
- in title? In email body?

Training

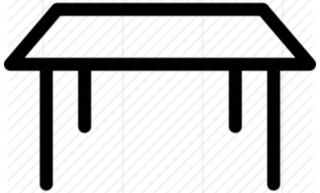


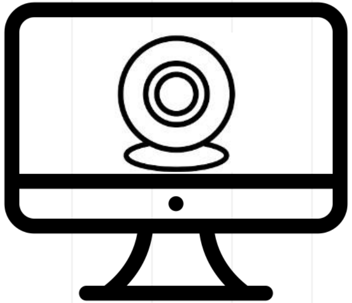
Prediction

What is Machine Learning?

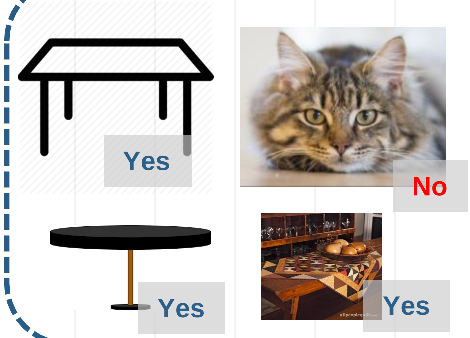
- **Example 2: Image Recognition**

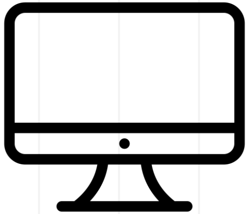
  Task: whether there is a table in the image





A table detector based on machine learning





conditions:

Flat Surface?

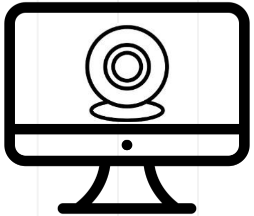
Legs or Support?

Height Consistency?

Edges and Corners?

Shadows and Reflections?

Training



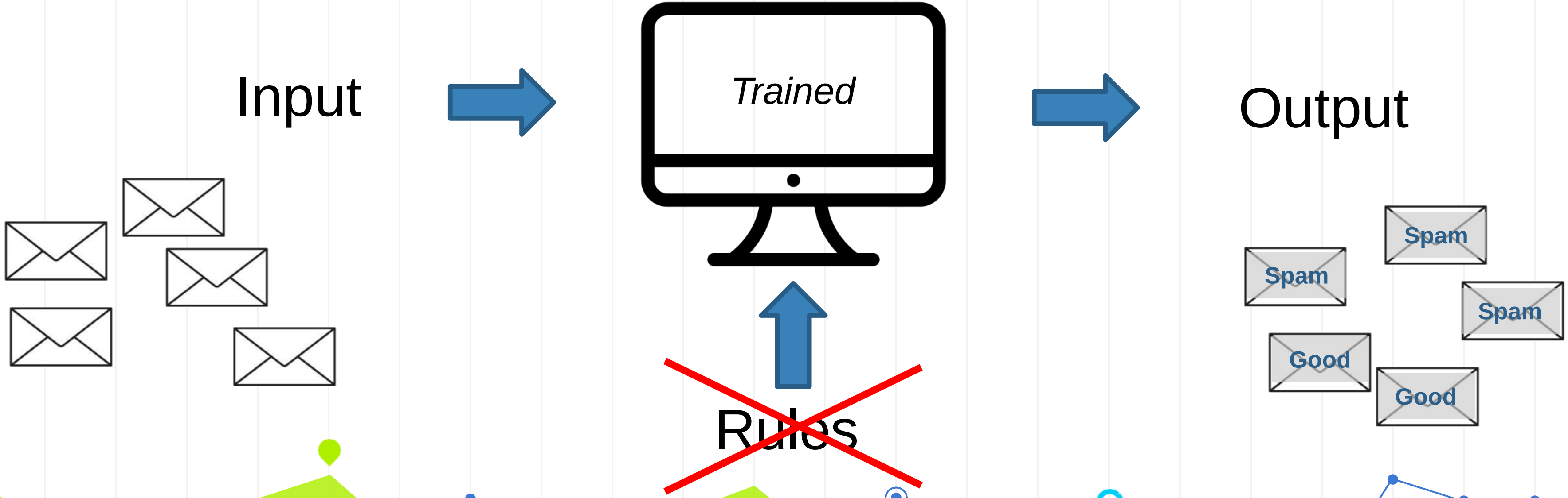
Yes

No

Prediction

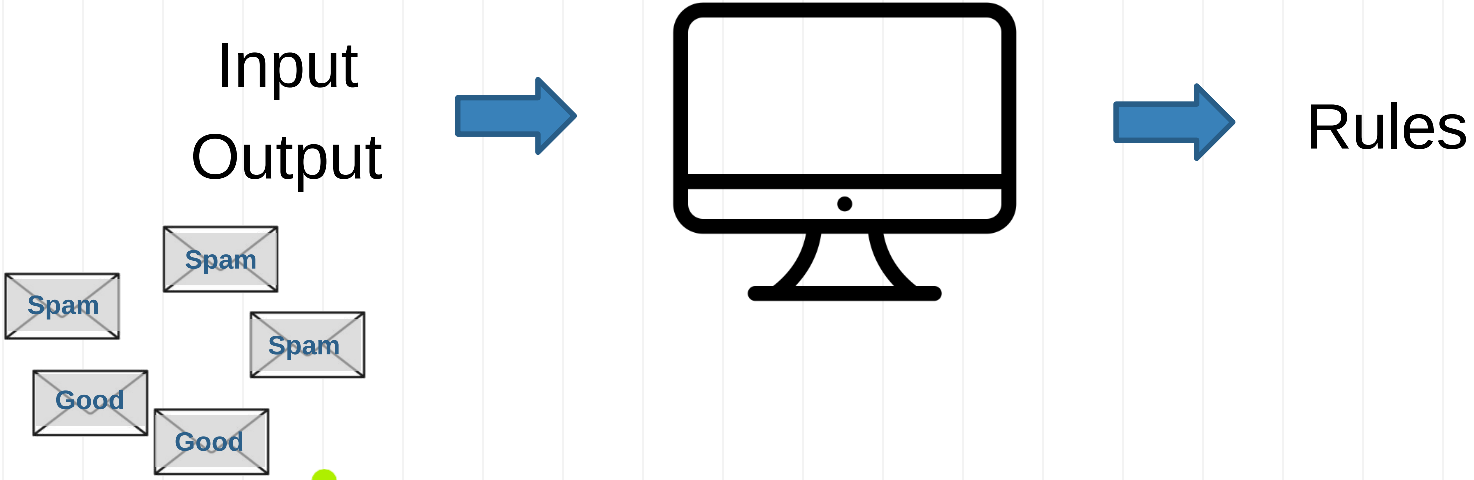
What is Machine Learning?

- We want a machine which can predict output accurately (most of time) based on input data.



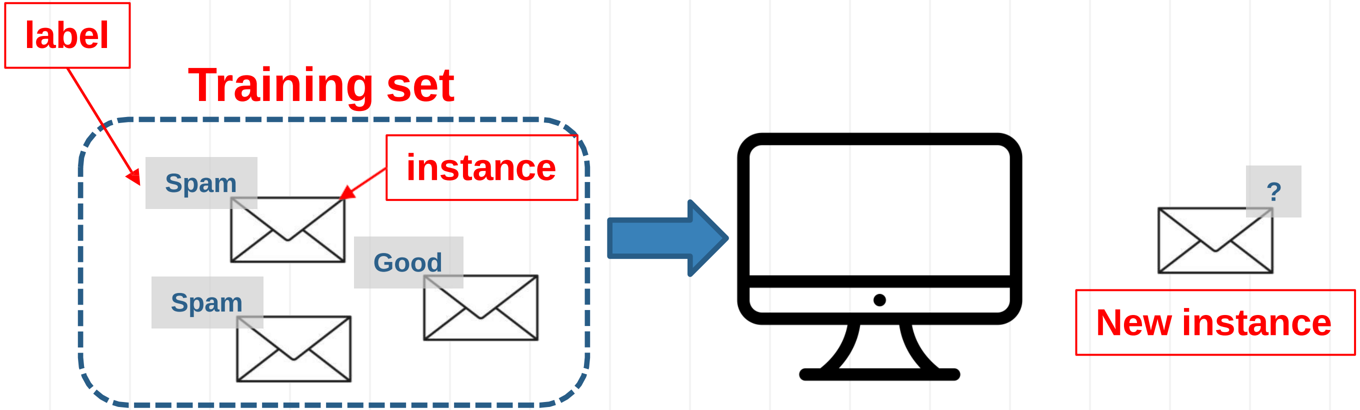
What is Machine Learning?

- Training



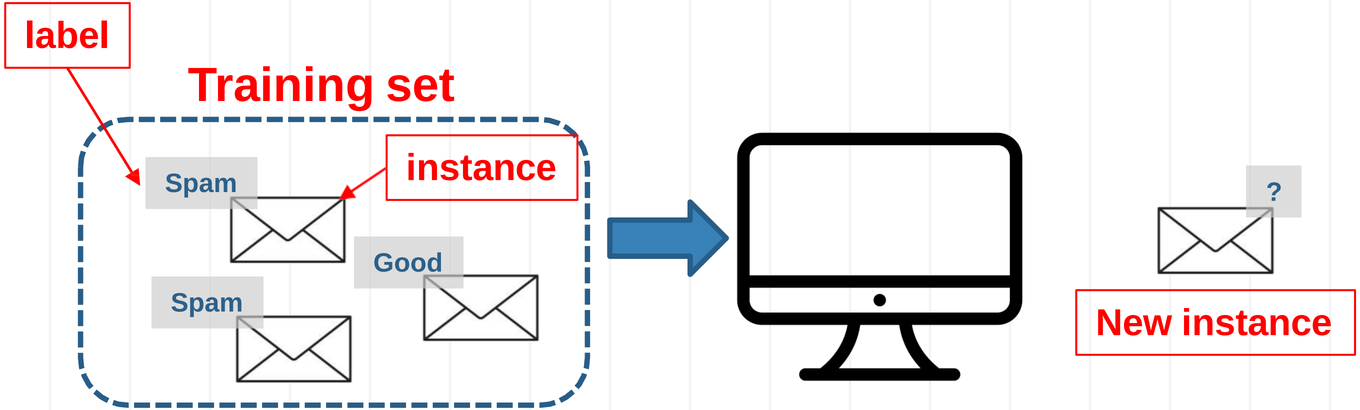
What is Machine Learning?

- Training



Supervised Learning

- Training



1. Supervised Learning: machine is trained with human supervision with a “teacher”, (the training set is labeled)

Then, what is unsupervised learning?

1. Supervised Learning: machine is trained with human supervision with a "teacher", (the training set is labeled)

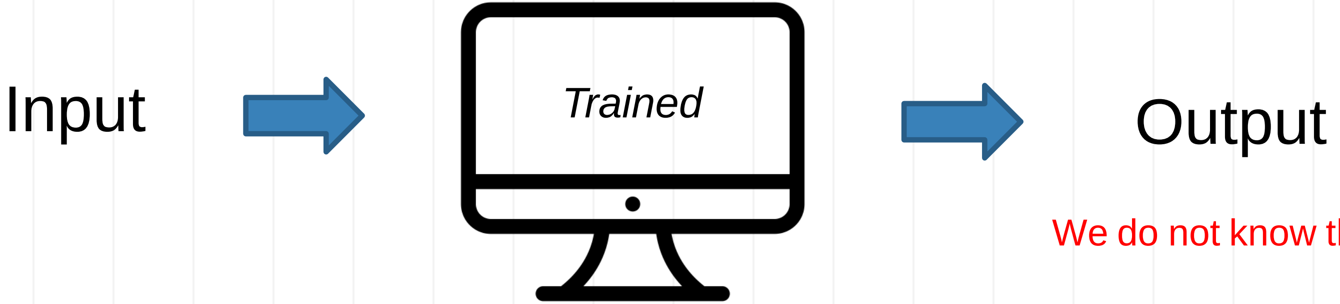
2. Unsupervised Learning: machine is trained without human supervision without a "teacher", (the training set is not labeled)

Unsupervised Learning

- We want to classify customers into different types based on their attributes

Unsupervised Learning

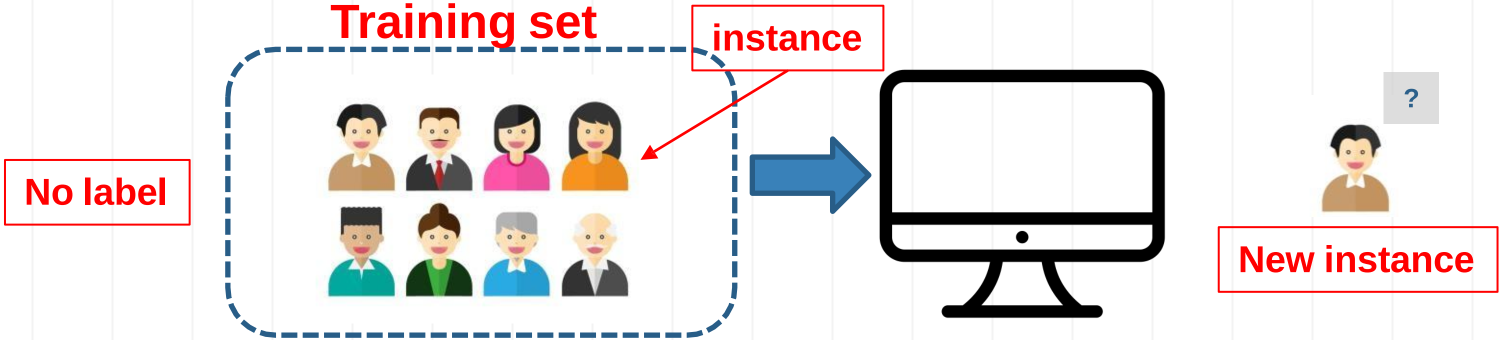
- We want to classify customers into different types based on their attributes
- Clustering data based on identified patterns or structures.



We do not know the output!

Unsupervised Learning

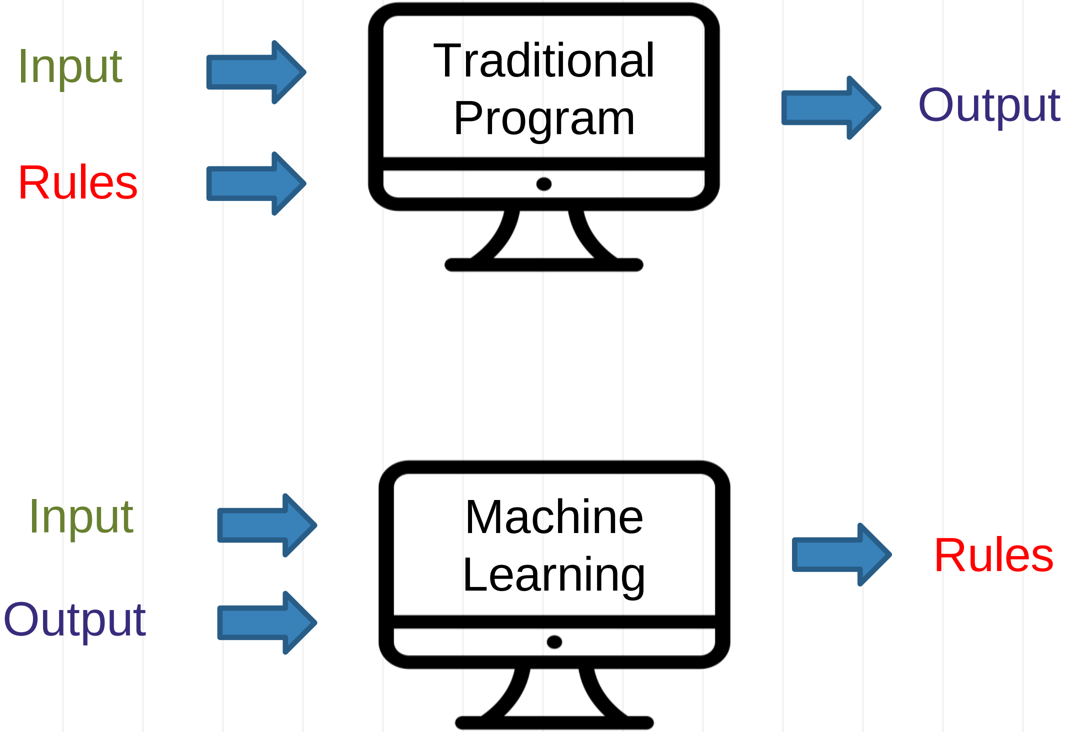
- Training



2. Unsupervised Learning: machine is trained without human supervision without a “teacher”, (the training set is not labeled)

Machine Learning Types

1. Supervised Learning: machine is trained with human supervision with a "teacher", (the training set is labeled)
2. Unsupervised Learning: machine is trained without human supervision without a "teacher", (the training set is not labeled)
3. Semisupervised Learning
4. Reinforcement Learning


Input
Rules
Traditional Program
Output
Input
Output
Machine Learning
Rules

Understanding Machine Learning

2

Understand Machine Learning

- What do I mean if I want to build a machine learning model for my housing transaction dataset?

Input

Housing datasets with attributes

SDIST	FAREA	BED	BATH
624.38916	86.71	1	1

Output

Housing prices

DTUP
40886.220

Understand Machine Learning

target

Output

Input

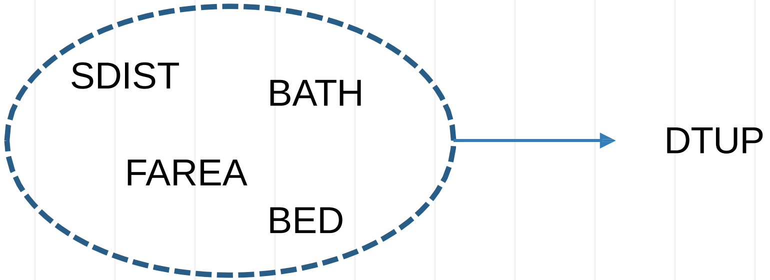
features

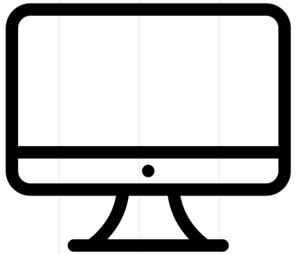
	DTUP	SDIST	FAREA	BED	BATH
1	40886.220	558.33545	131.37	3	2
2	27734.275	603.46985	59.59	2	1
3	28393.690	865.78906	48.02	1	1
4	33236.010	340.77704	135.75	3	2
5	33183.953	2037.08520	87.53	2	1
...	...	...	...	...	...
996	45743.414	624.38916	86.71	1	1
997	34796.133	282.94788	66.02	1	1
998	29992.648	386.15970	85.41	2	1
999	70583.040	526.79364	72.00	3	1
1000	42302.560	577.59314	109.00	2	1

Features, also known as attributes or variables, are the individual measurable characteristics of the data that are used as input.

The **target**, also known as the label or dependent variable, is the outcome or value that the machine learning model is trying to predict.

Understand Machine Learning





- How about a linear regression model?
  - Linear regression is a type of machine learning model

$$\boldsymbol{DTUP} = \beta_0 + \beta_1 \boldsymbol{SDIST} + \beta_2 \boldsymbol{FAREA} + \beta_3 \boldsymbol{BED} + \beta_4 \boldsymbol{BATH} + \varepsilon$$

- Is a regression model a supervised model or an unsupervised model?

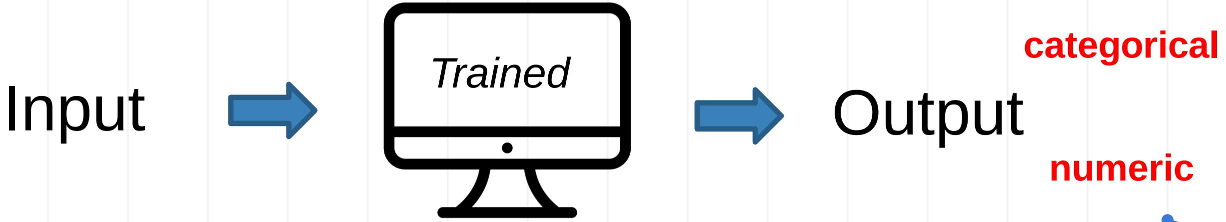
A regression model is a supervised learning model.

Supervised Learning

- **Two tasks :**
  - Classification: output is **categorical**
    - Spam email or not
    - With a table or not
  - Regression: output is **numeric**
    - Housing price

Logistic Regression is here!!!

Linear Regression is here!!!



Supervised Learning

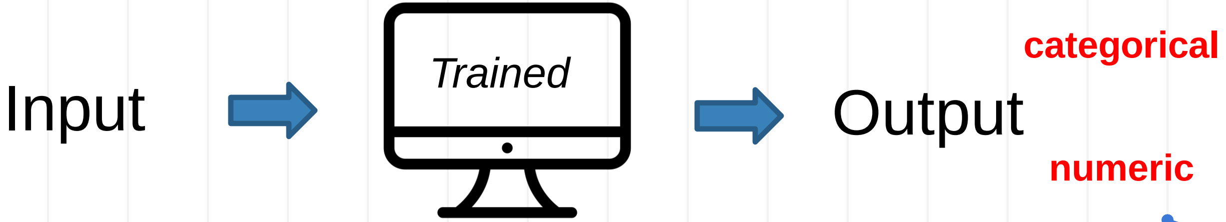
- **Two tasks :**

  - Classification: output is **categorical**
    - Spam email or not
    - With a table or not

  - Regression: output is **numeric**
    - Housing price

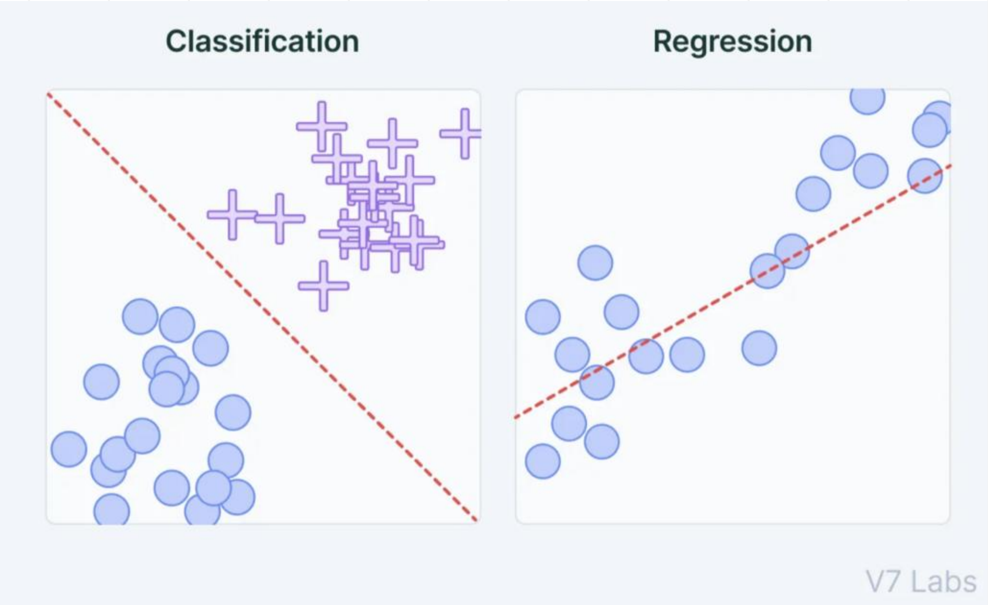
Logistic Regression is here!!!
Decision Trees; Random Forest; Support Vector Machines (SVM); k-Nearest Neighbors (k-NN)

Linear Regression is here!!!
Polynomial Regression
Ridge Regression / Lasso Regression
XGBoost



Supervised learning: Classification

- Here, we have a two-dimensional (read: two columns) dataset
- We know which dots are blue and which are red.
- The classification question then asks:
    - Can we create a “separator” model that separates these two points?



Which is classification problem?

- An e-commerce company using labeled customer data to predict whether or not a customer will purchase a particular item

- A restaurant using review data to ascribe positive or negative sentiment to a given review

- A bike share company using time and weather data to predict the number of bikes being rented at any given hour

Which is classification problem?

- An e-commerce company using labeled customer data to predict whether or not a customer will purchase a particular item

- A restaurant using review data to ascribe positive or negative sentiment to a given review

- A bike share company using time and weather data to predict the number of bikes being rented at any given hour

Machine Learning vs. Statistics

$$\boldsymbol{DTUP} = \beta_0 + \beta_1\boldsymbol{SDIST} + \beta_2\boldsymbol{FAREA} + \beta_3\boldsymbol{BED} + \beta_4\boldsymbol{BATH} + \varepsilon$$

Machine learning cares about prediction

- field of predictive modeling
- concerned with minimizing the prediction error or making the most accurate predictions
- borrow algorithms from statistics for predication purposes

Statistics cares about estimation and inference

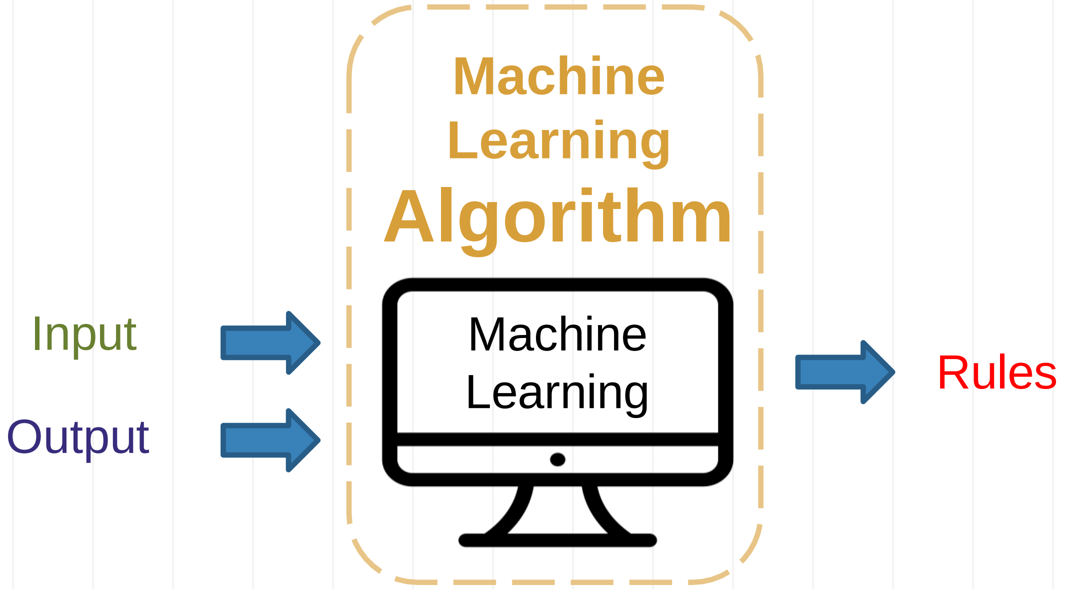
- field of statistical modeling
- understanding the relationship between variables
- many models/algorithms have been borrowed by machine learning.

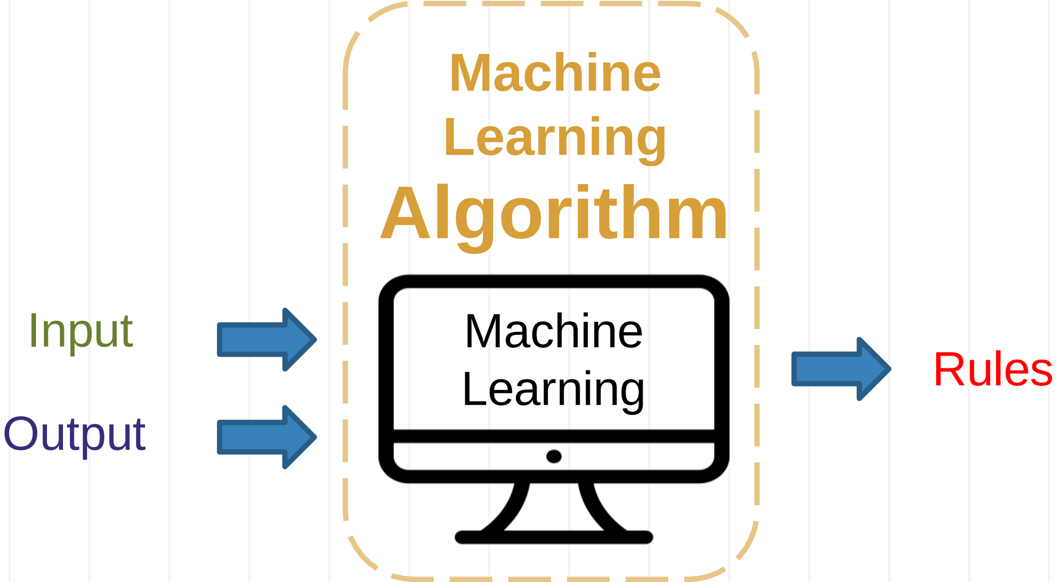
Machine Learning vs. Statistics

- **Statistics**: hypothesis testing, model assumptions, explanation, and interpretation

- In **machine learning**, different than in applied statistics, we are less interested in what these parameters are, and more in how well they can
  - Make **predictions**
  - Describe **underlying structures or characteristics in the data**

Machine Learning Algorithm 3


Machine Learning Algorithm
Input
Output
Machine Learning
Rules



Algorithm

- In mathematics and computer science, an **algorithm** is a finite sequence of well-defined, computer-implementable instructions, typically to solve a class of problems or to perform a computation. (Wikipedia).

- A **machine learning algorithm** is a special type of algorithm that learns from data instead of following fixed rules. It adjusts itself by analyzing input-output pairs and discovers patterns or decision rules **automatically**.

Machine Learning Algorithm

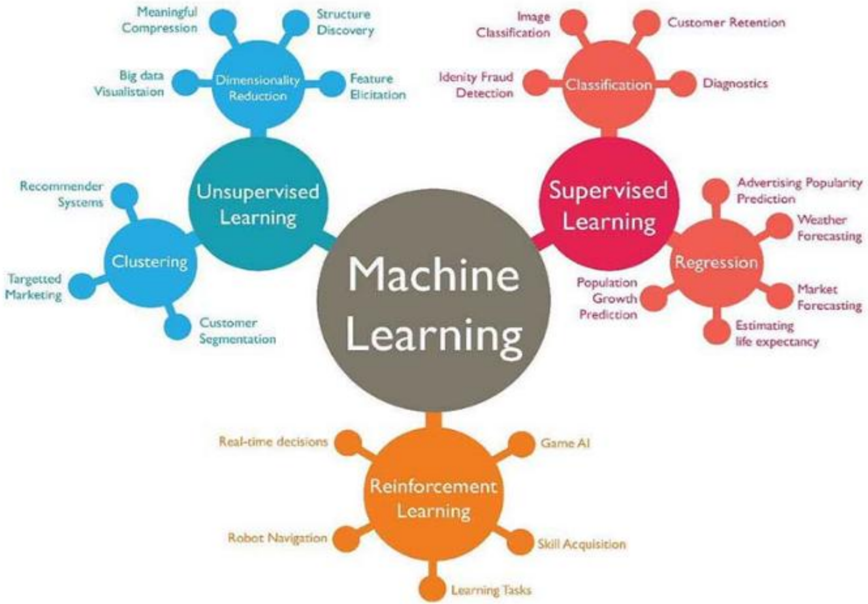
- **Supervised learning**
  - Classification
    - ❑ K-nearest neighbors
    - ❑ logistic regression
    - ❑ Random forest

      …
  - Regression
    - ❑ Linear regression

      …
- **Unsupervised learning**
  - Clustering
    - ❑ K-means clustering

      …
  - Dimensionality Reduction
    - ❑ PCA and factor analysis

      …



Algorithm Example: Linear Regression

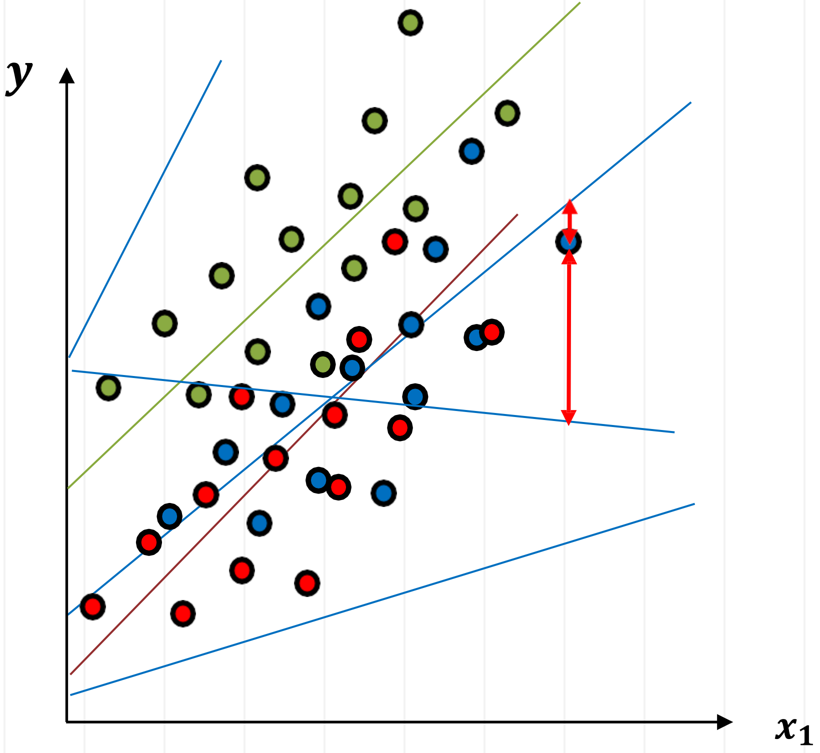
- **Supervised learning**
  - Regression Problem

y	x_1
20000	800
60000	2000
14000	632
54000	1800
20000	750
20000	400
…	…

$$y_i = \beta_0 + \beta_1 x_{1i} + \varepsilon_i$$

OLS: Ordinary Least Squares

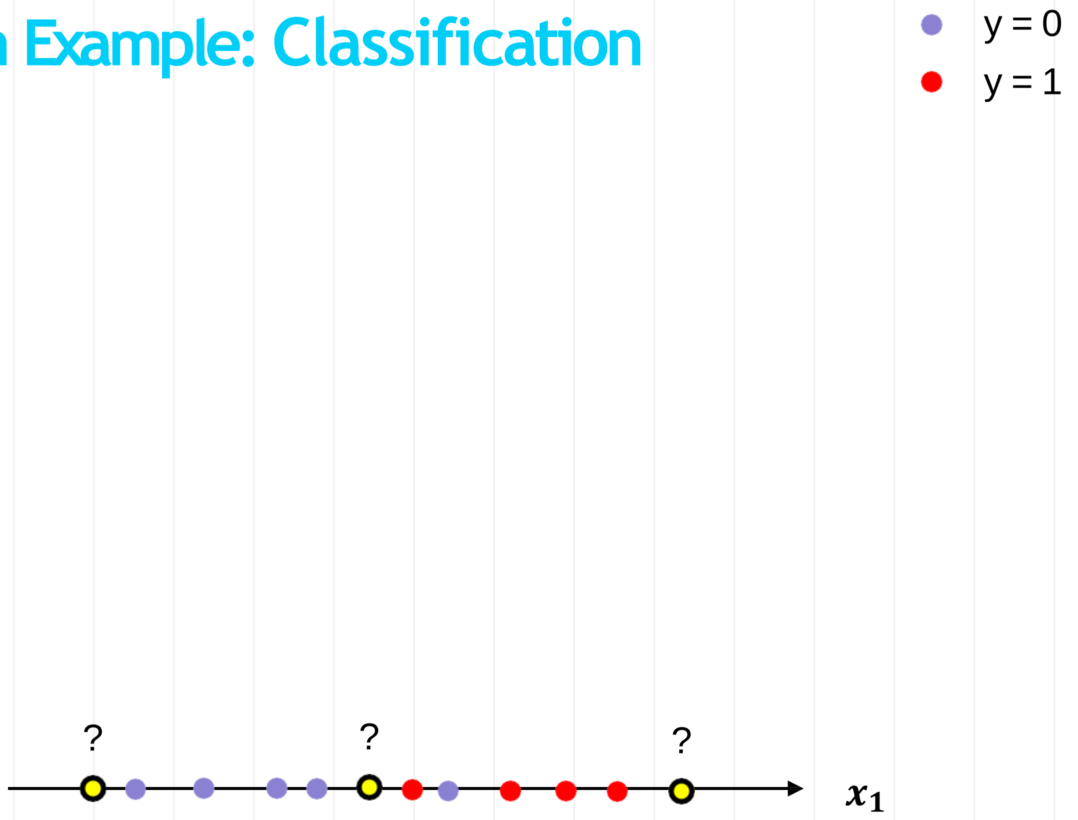
minimize sum of squared errors



Algorithm Example: Classification

- **Supervised learning**
  - Classification Problem
    - Binary Choice

y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…



Algorithm Example: Logistic Regression

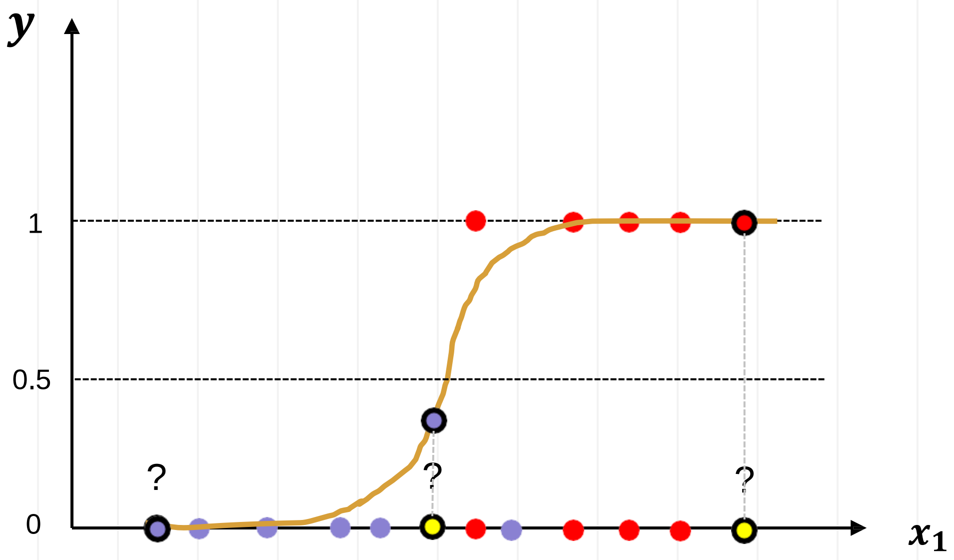
y = 0

y = 1

- **Supervised learning**
  - Classification Problem
    - Binary Choice

y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…

$$P(y_i = 1|x_{1i}) = \frac{\exp(\beta_0 + \beta_1 x_{1i})}{1 + \exp(\beta_0 + \beta_1 x_{1i})}$$

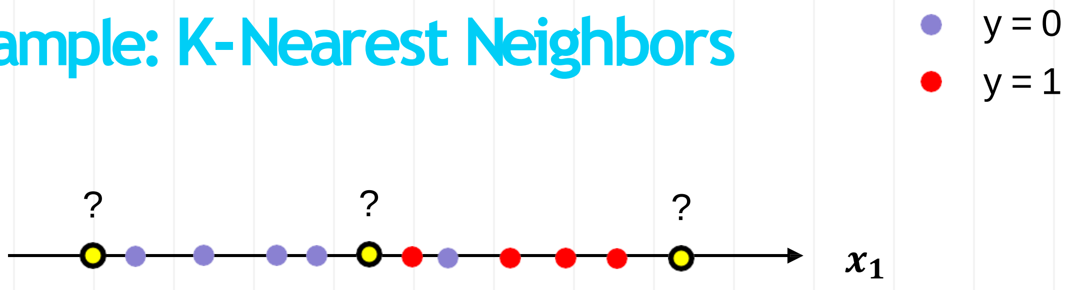


MLE: Maximum Likelihood Estimation

Algorithm Example: K-Nearest Neighbors

- **Supervised learning**
  - Classification Problem
    - Binary Choice

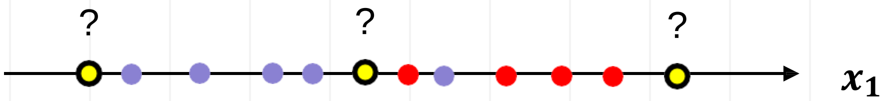
y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…



Algorithm Example: K-Nearest Neighbors

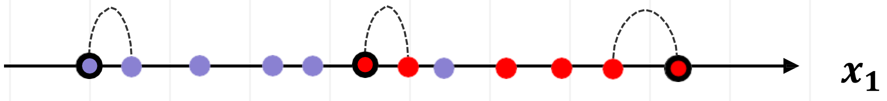
- y = 0
- y = 1

- **Supervised learning**
  - Classification Problem
    - Binary Choice

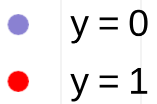


y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…

1 Nearest Neighbor

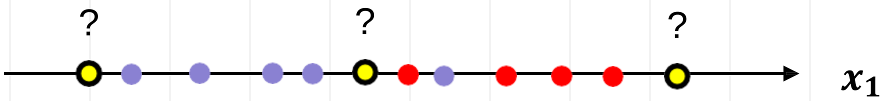


Algorithm Example: K-Nearest Neighbors

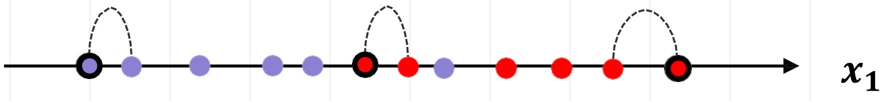


- **Supervised learning**
  - Classification Problem
    - Binary Choice

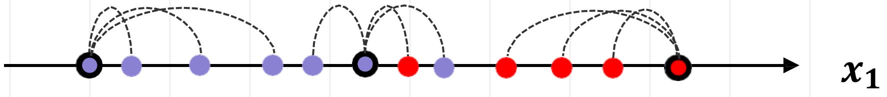
y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…



1 Nearest Neighbor



3 Nearest Neighbor



k Nearest Neighbor…

Algorithm Example: K-Nearest Neighbors

- **Supervised learning**
  - Classification Problem
    - Binary Choice

y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…

KNN is often referred to as a "lazy learner“

•it simply stores the training data and makes predictions by calculating distances to the known data points

The number of nearest neighbors (k) to consider when making a prediction

•**Small k:** Can lead to a model that is sensitive to noise in the training data (overfitting).

•**Large k:** Can lead to a model that is too generalized (underfitting).

4

Machine Learning Steps

Machine Learning Steps

- Gathering and loading data (what features to collect)
- Exploring data (e.g., pandas and visualization)
- Transforming data (e.g., string to numeric)
- Splitting data for training and testing
- Choosing and creating a model
- Training
- Testing (evaluating accuracy)
- Tuning the model (hyperparameters)
- Making predictions on new data

Data Preparation

Transforming data (e.g., string to numeric)

Label Encoding: Converts categorical labels into numerical values.
["red", "green", "blue"] -> [0, 1, 2]

One-Hot Encoding: Converts categorical labels into numerical values.
["red", "green", "blue"] -> three dummy variables

Ordinal Encoding: Similar to label encoding but used for ordinal categories where there is an inherent order.
["low", "medium", "high"] -> [1, 2, 3]

Splitting data for training and testing

Training set

75%

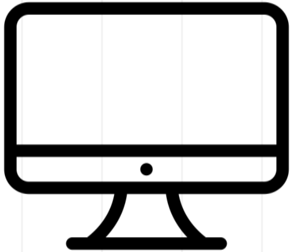
(80%)

Testing set

25%

(20%)

DTUP	SDIST	FAREA	BED	BATH
40886.220	558.33545	131.37	3	2
27734.275	603.46985	59.59	2	1
28393.690	865.78906	48.02	1	1
33236.010	340.77704	135.75	3	2
33183.953	2037.08520	87.53	2	1
...	...	...	...	...
45743.414	624.38916	86.71	1	1
34796.133	282.94788	66.02	1	1
29992.648	386.15970	85.41	2	1
70583.040	526.79364	72.00	3	1
42302.560	577.59314	109.00	2	1



Machine Learning Steps

- Gathering and loading data
- Exploring data (e.g., pandas and visualization)
- Transforming data (e.g., string to numeric)
- Splitting data for training and testing
- Choosing and creating a model
- Training
- Testing (evaluating accuracy) How to measure accuracy?
- Tuning the model (hyperparameters)
- Making predictions on new data

Classification (Supervised Learning)

- **Example**: Spam Email Filter

- We prepared a dataset with 5000 emails (with features and labels)
- We split the dataset into training set (4000, 80%) and testing set (1000, 20%)
- We created a model (e.g., k-nearest neighbors or logistic regression)
- We trained the model using the training set (4000 instances)
- Now, we want to test the model and evaluate the accuracy…

Classification (Supervised Learning)

- **Example:** Spam Email Filter
- Now, we want to test the model and evaluate the accuracy…
    - We predict the labels of the 1000 instances in the testing set
    - Then compare with their actual labels

Actual: [spam, spam, spam, good, ..., good, good, good]

Predicted: [spam, good, good, spam, ..., spam, good, good]

		Predicted	
		Spam	Good
Actual	Spam		
	Good		

Confusion Matrix

- Describe the performance of a **classification** model

		Predicted	
	Total = 1000	**Spam**	**Non-Spam**
Actual	**Spam**	330	70
	Non-Spam	90	510

		Predicted	
		Spam	**Non-Spam**
Actual	**Spam**	True Positive	False Negative
	Non-Spam	False Positive	True Negative

Type II error

Type I error

Overall, how often is the model correct?

Metric 1: Accuracy

$$Accuracy = \frac{True\ Positive + True\ Negative}{Total}$$

$$= \frac{330 + 510}{1000}$$

$$= 0.84$$

Confusion Matrix

How often does the model correctly identify positives (spam emails)?

		Predicted	
	Total = 1000	Spam	Non-Spam
Actual	Spam	330	70
	Non-Spam	90	510

		Predicted	
		Spam	Non-Spam
Actual	Spam	True Positive	False Negative
	Non-Spam	False Positive	True Negative

Metric 2: Recall

(Sensitivity or True Positive Rate)

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

$$= \frac{330}{330 + 70}$$

$$= 0.825$$

Confusion Matrix

When the model predicts positive, how often is it correct?

Metric 3: Precision

Predicted

Actual	Total = 1000	Spam	Non-Spam
	Spam	330	70
	Non-Spam	90	510

Predicted

Actual		Spam	Non-Spam
	Spam	True Positive	False Negative
	Non-Spam	False Positive	True Negative

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

$$= \frac{330}{330 + 90}$$

$$\approx 0.786$$

Confusion Matrix

How often does the model correctly identify negatives(non-spam emails)?

Total = 1000	Predicted: Spam	Predicted: Non-Spam
Actual: Spam	330	70
Actual: Non-Spam	90	510

	Predicted: Spam	Predicted: Non-Spam
Actual: Spam	True Positive	False Negative
Actual: Non-Spam	False Positive	True Negative

Metric 4: True Negative Rate (Specificity)

$$= \frac{True\ Negative}{True\ Negative + False\ Positive}$$

$$= \frac{510}{510 + 90}$$

$$= 0.85$$

Confusion Matrix

Predicted

Total = 1000	Spam	Non-Spam
Spam	330	70
Non-Spam	90	510

Actual

Predicted

	Spam	Non-Spam
Spam	True Positive	False Negative
Non-Spam	False Positive	True Negative

Actual

Other Metrics

Error Rate

$$= \frac{False\ Positive + False\ Negative}{Total}$$

False Positive Rate

$$= \frac{False\ Positive}{False\ Positive + True\ Negative}$$

More Than Two Classes

Total = 1000	A	B	C
A	280	8	12
B	15	260	25
C	30	50	320

Predicted (columns); Actual (rows)

Overall, how often is the model correct?

Metric 1: Accuracy

$$= \frac{280 + 260 + 320}{1000} = 0.86$$

More Than Two Classes

How often does the model correctly identify ~~positives (spam emails)?~~ each class

Metric 2: Recall

(Sensitivity or True Positive Rate)

Total = 1000	A	B	C
A	280	8	12
B	15	260	25
C	30	50	320

Predicted (columns); Actual (rows)

$$Recall_A = \frac{280}{280 + 8 + 12} \approx 0.93$$

$$Recall_B = \frac{260}{260 + 15 + 25} \approx 0.87$$

$$Recall_C = \frac{320}{320 + 30 + 50} = 0.8$$

More Than Two Classes

When the model predicts ~~positive~~, how often is it correct?
each class

Metric 3: Precision

Predicted

Actual	Total = 1000	A	B	C
	A	280	8	12
	B	15	260	25
	C	30	50	320

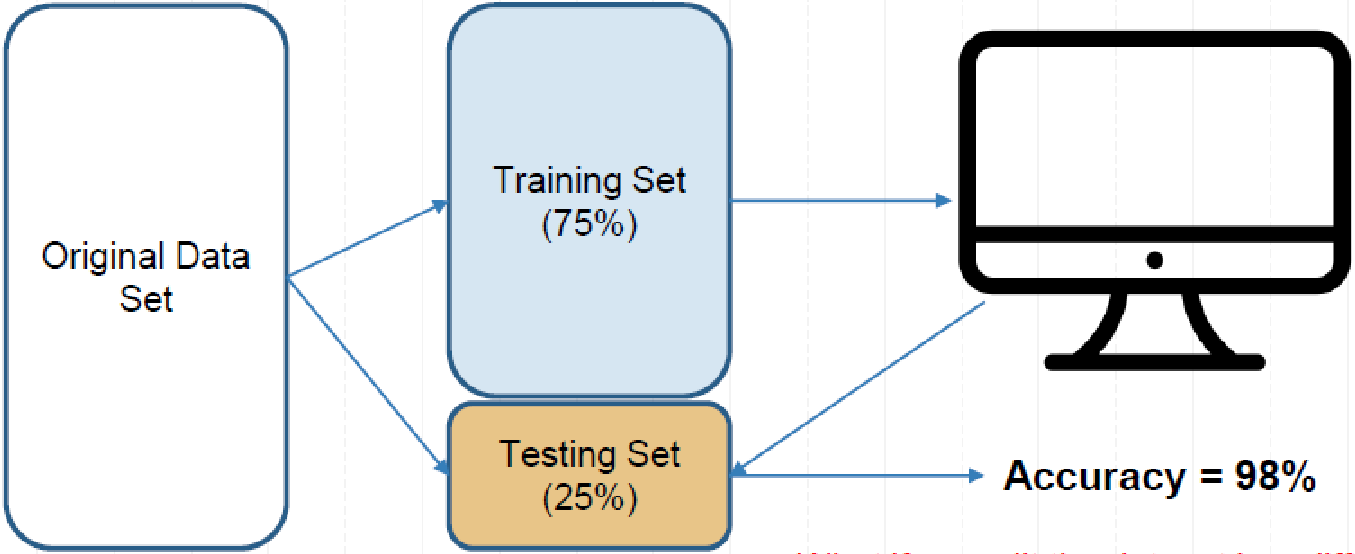
$$Precision_A = \frac{280}{280 + 15 + 30} \approx 0.86$$

$$Precision_B = \frac{260}{260 + 8 + 50} \approx 0.82$$

$$Precision_C = \frac{320}{320 + 12 + 25} \approx 0.90$$

Cross-Validation

Is the Model Ready for Use?



What if we split the dataset in a different way?

Should we find another dataset to test our model?

Cross-Validation

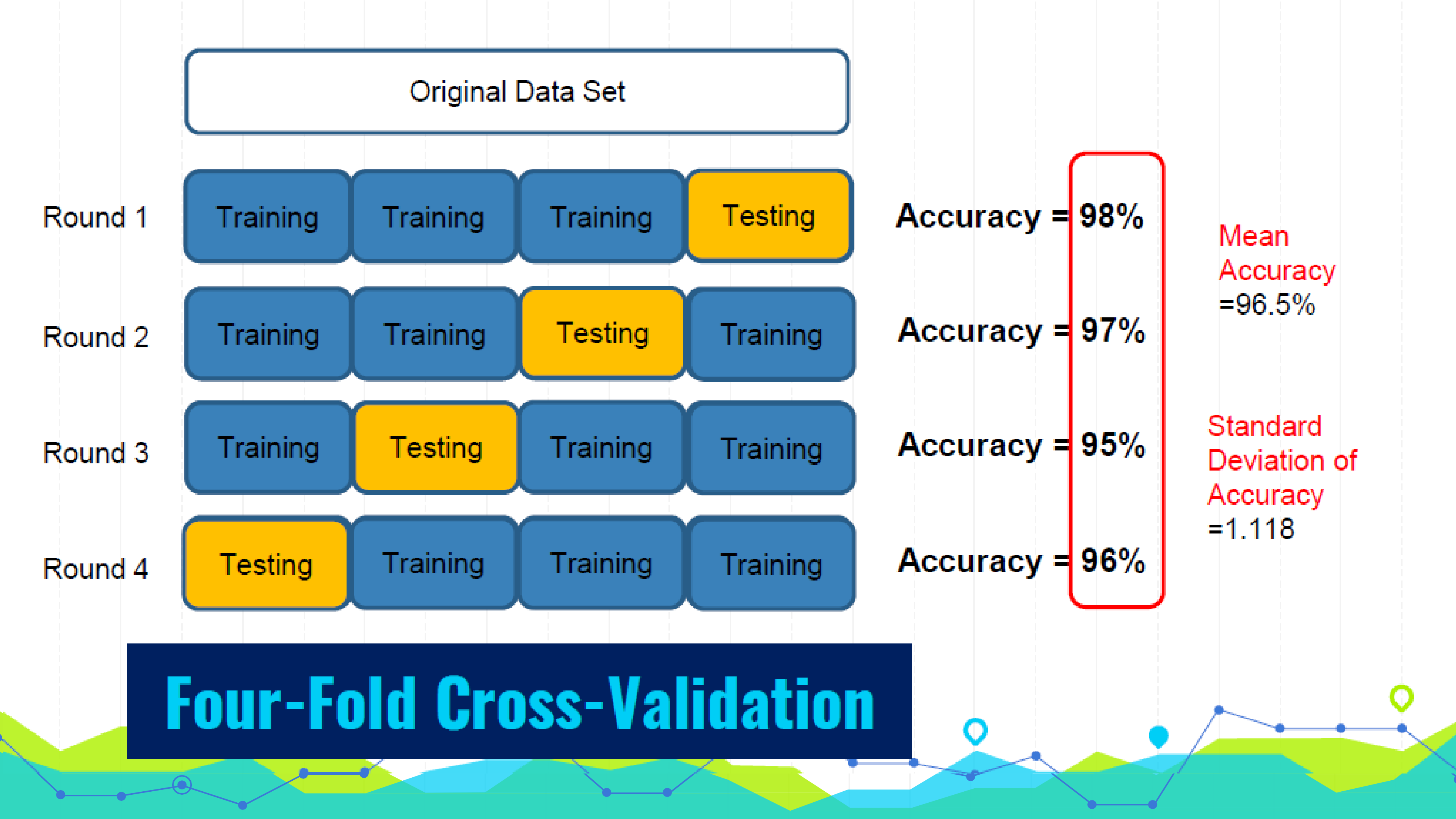
- **Cross-validation** is a resampling procedure used to evaluate machine learning models on a limited data sample.

1. Splitting the Data: The dataset is split into K equal-sized (or nearly equal-sized) subsets.

2. Training and Validation: The model is trained K times, each time using K-1 folds for training and the remaining 1 fold for validation.

3. Rotation: The validation fold is rotated such that each of the K folds is used exactly once as the validation set.

4. Averaging the Results: The performance metric (e.g., accuracy, precision, recall) is averaged across all K trials to give an overall performance estimate.


Original Data Set
Round 1
Training
Training
Training
Testing
Accuracy = 98%
Round 2
Training
Training
Testing
Training
Accuracy = 97%
Round 3
Training
Testing
Training
Training
Accuracy = 95%
Round 4
Testing
Training
Training
Training
Accuracy = 96%
Mean Accuracy =96.5%
Standard Deviation of Accuracy =1.118
Four-Fold Cross-Validation

Cross-Validation

- **K**-Fold Cross-validation
- **K** is the number of equal-size blocks you split the data into
- Ten-Fold Cross Validation is a common choice.

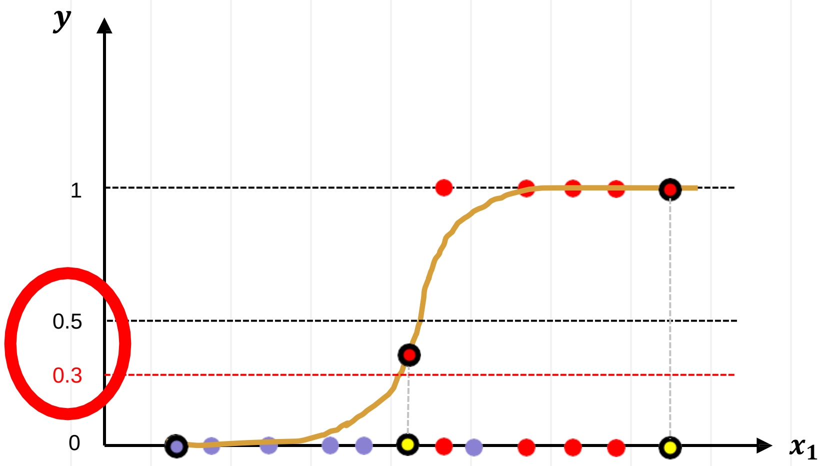
Machine Learning Steps

- Gathering and loading data
- Exploring data (e.g., pandas and visualization)
- Transforming data (e.g., string to numeric)
- Splitting data for training and testing
- Choosing and creating a model
- Training
- Testing (evaluating accuracy)
- Tuning the model (hyperparameters)
- Making predictions on new data

Hyperparameter Tuning

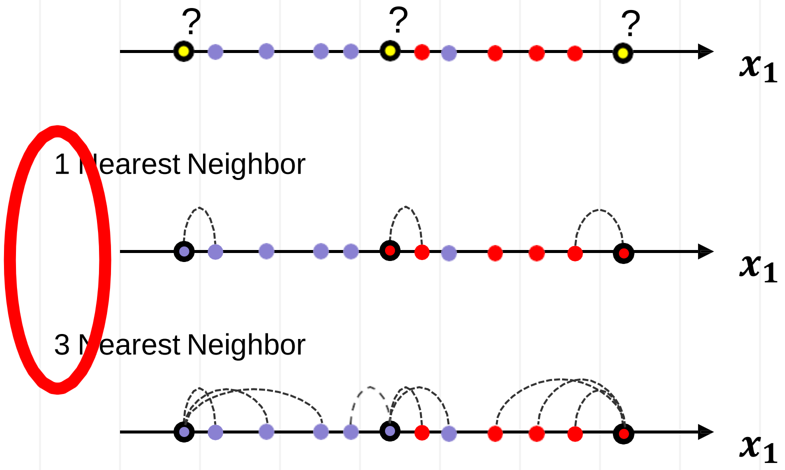
- **Hyperparameter**: model parameters specified in advance (before training)

What **threshold** to use?



Logistic Regression

What **k** to use?

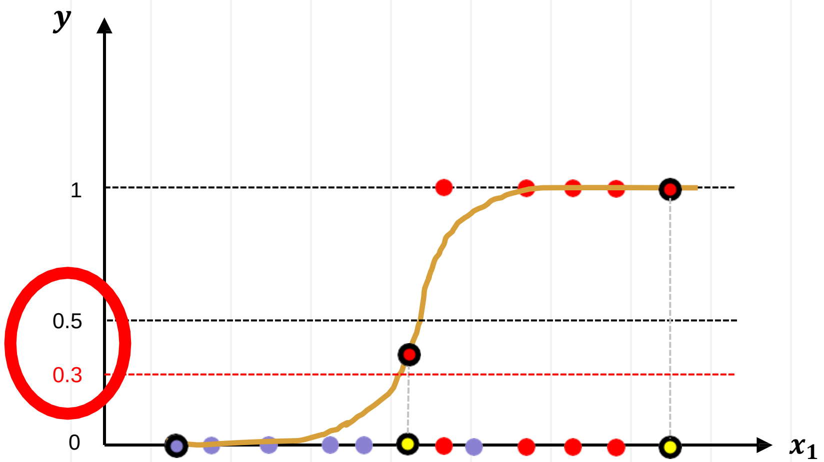


K-Nearest Neighbors

Hyperparameter Tuning

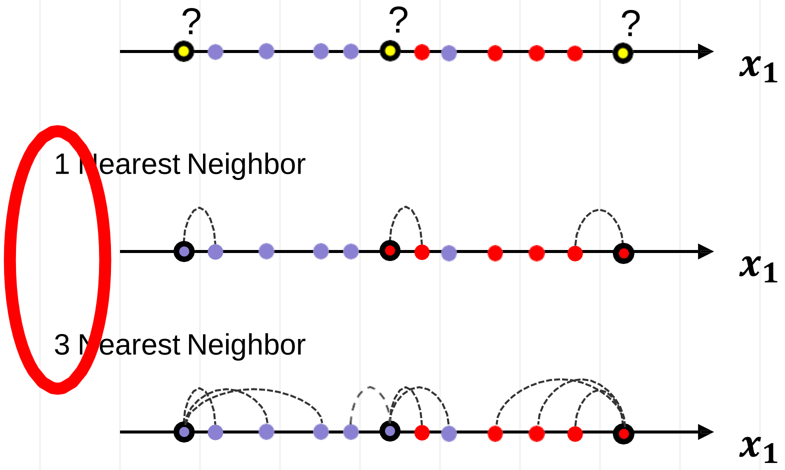
- **Hyperparameter**: model parameters specified in advance (before training)

What **threshold** to use?



Logistic Regression

What **k** to use?



K-Nearest Neighbors

When to Use

- Testing a model
- Hyperparameter Tuning
- Comparing models

Metrics for Accuracy

+

K-Fold Cross-Validation

Machine Learning Steps

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- Testing (evaluating accuracy)
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- Making predictions on new data

Q&A

Any questions?

You can find me at

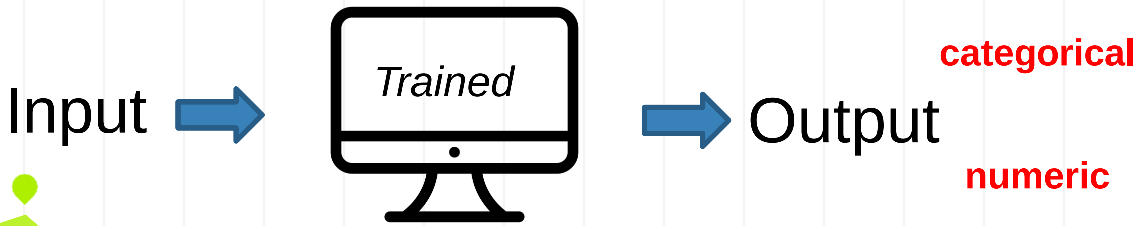
wl563@cornell.edu

Machine Learning is Great For

1. Problems for which transitional solutions require a lot of fine-tuning or long lists of rules
2. Complex problems for which the traditional approach yields no good solution
3. Fluctuating environments
4. Getting insights about complex problems and large amounts of data

Supervised Learning

- Associating features with some label.
- **Two tasks :**
  - Classification: labels as discrete categories
    - Spam email or not
    - With a table or not
  - Regression: labels as continuous quantities.
    - Housing price



Training

target

Output | Input | features

	DTUP	SDIST	FAREA	BED	BATH
1	40886.220	558.33545	131.37	3	2
2	27734.275	603.46985	59.59	2	1
3	28393.690	865.78906	48.02	1	1
4	33236.010	340.77704	135.75	3	2
5	33183.953	2037.08520	87.53	2	1
...	...	...	...	...	...
996	45743.414	624.38916	86.71	1	1
997	34796.133	282.94788	66.02	1	1
998	29992.648	386.15970	85.41	2	1
999	70583.040	526.79364	72.00	3	1
1000	42302.560	577.59314	109.00	2	1

Features, also known as attributes or variables, are the individual measurable characteristics of the data that are used as input.

The **target**, also known as the label or dependent variable, is the outcome or value that the machine learning model is trying to predict.